Locality-Based Explanations with LLMs: A Reflection on Human-Centered Prompt Design

Liziane Santos Soares[1,2], Lucas da Silva Santos[1], Maria Carvalhido Izabel Barreto[1], Jussara Marques de Almeida[1], Marcos André Gonçalves[1], Raquel Oliveira Prates[1]

[1]Universidade Federal de Minas Gerais – Belo Horizonte, Brazil

[2]Universidade Federal de Viçosa – Rio Paranaíba, Brazil

liziane.soares@dcc.ufmg.br,
{lucsantslucas,mariacarvalhido}@ufmg.br,
{jussara, mgoncalv, rprates}@dcc.ufmg.br

***Abstract. Introduction**: Locality-based text classification approaches in machine learning rely on semantic neighborhoods that may be computationally coherent while remaining difficult for humans to interpret. Large Language Models (LLMs) create opportunities to transform these locality structures into natural-language explanations, but also introduce new human-centered challenges related to ambiguity, uncertainty communication, and explanation literacy. **Objective**: This paper presents a reflective account of designing LLM-mediated locality explanations, investigating how prompt design can support interpretability across audiences with different levels of technical expertise while preserving structural aspects of the locality. **Methodology**: Grounded in HCI and XAI literature, we conducted an iterative and human-guided prompt design process involving multiple refinement cycles across different LLMs, neighborhood configurations, and audience profiles. The process focused on identifying and mitigating recurrent failure patterns, including topic dominance, hallucinated coherence, and causal over-justification. **Results**: This work contributes emerging design insights for human-centered locality explanations, including audience-aware prompting strategies and communicative interventions aimed at preserving ambiguity and semantic variability in generated explanations. Beyond the specific context of locality-based explanations, the resulting insights may contribute to the development of explainability approaches in domains where complex technical information must be communicated across different levels of abstraction and user expertise.*

***Keywords** Explainable AI, Locality-Based Explanations, LLM-Generated Summaries, Human-Centered Prompt Design*

1. Introduction

In the context of Explainable Artificial Intelligence (XAI) for Automatic Text Classification (ATC), locality-based approaches use semantically similar texts as contextual evidence to help users understand why a particular classification was produced, rather than relying exclusively on internal model behavior or features of the classified instance itself [Soares et al. 2026]. In this context, *locality*, also referred to as a *neighborhood*, corresponds to the local semantic structure surrounding a classified

instance, typically represented by semantically similar texts in the representation space [Molnar 2020]. Several XAI approaches for ATC rely on locality-based strategies to support the interpretation of model decisions, including techniques such as LIME and SHAP [Ribeiro et al. 2016, Saito et al. 2020, Ghalebikesabi et al. 2021, Gaudel et al. 2022, Lundberg and Lee 2017].

Although locality structures may be computationally coherent, they are not inherently legible to human users. Neighborhoods may exhibit semantic cohesion, topic overlap, ambiguity, or class dispersion, making their interpretation cognitively demanding, particularly for non-technical audiences. Large Language Models (LLMs) create new opportunities to transform these locality structures into natural-language explanations grounded in neighborhood evidence [Soares et al. 2026]. However, this transformation introduces a fundamentally human-centered challenge: explanations should not merely sound coherent, but should reliably communicate the structural properties and uncertainty patterns of the neighborhood itself.

This challenge becomes particularly relevant when explanations must be adapted to users with different levels of AI and data literacy. Technical audiences may benefit from explicit representations of semantic structure, uncertainty, and class dispersion, whereas non-technical users often require explanations that prioritize clarity, contextualization, and communicative accessibility [Kim et al. 2024]. In this sense, prompting strategies are not merely mechanisms for controlling LLM outputs. They also function as communicative design instruments that mediate how AI explanations are interpreted across different audiences.

Where visual explanation metaphors such as word clouds emphasize the most frequent or salient terms within local neighborhoods [Hicke et al. 2022, Colonel and Lin 2025], LLM-generated locality narratives create opportunities to communicate nuanced semantic patterns, dominant themes, ambiguity, and local evidence through natural language explanations. These explanation strategies align with ongoing discussions on *data* and *AI literacy* within Human-Centered AI (HCAI) [Kim et al. 2024] and Human-Data Interaction (HDI) [Mortier et al. 2014], which emphasize the user's ability to critically interpret and engage with AI-generated information and explanations [Coleti et al. 2024, Pereira et al. 2025, Ge et al. 2026].

At the same time, locality summarization differs substantially from conventional Natural Language Processing (NLP) summarization tasks [Zhang et al. 2025a, Zhang et al. 2024, Fonseca and Cohen 2024]. Prompting strategies optimized purely for fluent summarization may inadvertently impose an artificial coherence, suppress uncertainty cues, attenuate semantic variability, or generate persuasive narratives that fail to accurately reflect heterogeneous local structures.

Adopting this HCI perspective, this paper presents a reflective account of designing LLM-mediated locality explanations through iterative, human-centered prompt design. Grounded in HCI and XAI literature, we conducted multiple cycles of prompt refinement across different LLMs, neighborhood configurations, and audience profiles. Rather than proposing a finalized summarization pipeline, our primary contribution lies in the emerging design insights derived from adapting and combining existing prompting strategies to the communicative needs of locality-based explanations. Specifically, we

address the following research question: *How can human-centered prompt design support the interpretability of LLM-mediated locality explanations across different levels of user expertise while reliably preserving the structural properties of the neighborhood?*

The remainder of this paper is organized as follows. Section 2 presents the Background and Related Work. Section 3 describes the design problem and the methodological approach adopted in this study. Section 4 discusses the iterative human-centered prompt design process and the main design challenges identified throughout the refinement cycles. Section 5 presents the reflective inspection of the generated summaries and the emerging insights obtained from this exploratory evaluation. Finally, Sections 6 and 7 discuss the study's contributions, limitations, and future research directions.

2. Background and Related Work

2.1. Locality and Explainability in Automatic Text Classification

The widespread adoption of Machine Learning (ML) systems has increased concerns about how people understand, interpret, and engage with algorithmic decisions. In response, Explainable Artificial Intelligence (XAI) research has advanced approaches aimed at improving transparency, explainability, and accountability [Tolomei et al. 2017, Das et al. 2020, Vilone and Longo 2021].

Explainability and interpretability are core concepts associated with making model behavior and results understandable [Mohseni et al. 2021, Zhou et al. 2021]. However, some studies distinguish interpretability as an intrinsic property of inherently comprehensible models, whereas explainability refers to post-hoc explanations generated for opaque systems [Marcinkevičs and Vogt 2023, Ali et al. 2023]. Following this distinction, the present work is situated within the scope of explainability for Automatic Text Classification (ATC).

In this context, *locality*, also referred to as a *neighborhood*, corresponds to the semantic structure surrounding a classified instance, typically represented by semantically similar texts in the representation space [Molnar 2020]. This concept underlies classical Machine Learning approaches such as k-nearest neighbors and relates to the clustering hypothesis in Information Retrieval, which states that closely associated documents tend to satisfy similar information needs [van Rijsbergen 1979]. Locality also plays an important role in XAI. Several locality-based approaches use neighboring instances to support the interpretation of model decisions, including techniques such as LIME and SHAP [Ribeiro et al. 2016, Saito et al. 2020, Ghalebikesabi et al. 2021, Gaudel et al. 2022, Lundberg and Lee 2017].

Recent studies suggest that the quality of locality-based explanations is strongly influenced by the structural properties of the locality itself. In particular, *locality homogeneity* has emerged as an important factor for explainability. It is commonly characterized through the semantic similarity and class-distribution entropy of the closest instances (texts). In this context, we define [Soares et al. 2026]:

- *Similarity*: the degree of semantic proximity among neighboring texts. Higher similarity indicates that neighboring instances discuss closely related topics and share more consistent semantic patterns.

- *Class-distribution entropy*: the degree of mixture or diversity of classes represented in the neighboring texts. Lower entropy indicates a more consistent neighborhood, with texts predominantly associated with the same classes. Higher entropy indicates a greater presence of conflicting topics, classes, or evidence within the locality, potentially generating uncertainty regarding a dominant theme.

Highly homogeneous localities (*high similarity* and *low entropy*) tend to exhibit concentrated semantic patterns and clearer explanatory signals. In contrast, *heterogeneous localities* (*low similarity* and *high entropy*) may contain overlapping topics, diffuse class boundaries, or conflicting evidence, making interpretation more difficult [Soares et al. 2026]. Although locality has been widely explored in XAI, existing approaches fail to explain how local semantic features, noise, and conflicting evidence are structured within the neighborhoods, leaving a critical knowledge gap regarding how these complex spaces are perceived, interpreted, and cognitively processed by human users in real-world explainability contexts.

2.2. From Visual Explanations to LLM-Mediated Explanations

The effectiveness of an explanation depends not only on its technical quality, but also on users' ability to interpret and meaningfully engage with the explanatory content [Hoffman et al. 2023, Miller 2019]. Although faithful explanations are important [Jacovi and Goldberg 2020, Li et al. 2024], their usefulness is ultimately shaped by how users comprehend, contextualize, and critically interpret the information being presented. In visual explanation contexts, these interpretive skills are commonly associated with *Data Visualization Literacy*, which refers to users' ability to decode and understand visual representations through processes of perception, interpretation, and comprehension [Rodrigues et al. 2024].

As AI systems become increasingly embedded in domains such as healthcare, education, and public services, important challenges emerge regarding users' ability to critically engage with data-driven systems. Educational barriers, digital inequalities, and overly technical forms of communication may limit users' autonomy and participation, particularly among non-technical audiences. Within the field of Human-Data Interaction (HDI), these concerns reinforce the importance of *Data Literacy*. In this context, users should be able not only to access algorithmic information, but also to critically interpret and meaningfully use it. Transparency, therefore, cannot be reduced to merely exposing technical details. Algorithmic processes must also remain understandable and accessible across different cognitive and social contexts [Coleti et al. 2024, Ge et al. 2026].

These concerns are closely related to the concept of *Explanation Literacy*, which refers to users' skills to interpret, critically assess, and appropriately use algorithmic evidence communicated through AI-generated explanations. While prior literacy research has predominantly focused on visual representations [Rodrigues et al. 2024], the increasing adoption of LLM-mediated explanations introduces new interpretative demands associated with natural-language narratives. These demands include ambiguity management, uncertainty communication, persuasive fluency, and contextual grounding [Shorinwa et al. 2025]. Consequently, explanations do not become inherently human-centered simply because they are expressed in fluent natural language.

2.3. Locality Summarization as a Human-Centered Design Challenge

Recent HCI discussions argue that trustworthy AI should not merely increase user trust, but instead support calibrated trust aligned with the system's actual capabilities and limitations [Beltrão et al. 2026]. This concern is particularly relevant for LLM-mediated explanations, whose linguistic fluency and narrative coherence may inadvertently promote overtrust when uncertainty and semantic ambiguity are insufficiently communicated [Shorinwa et al. 2025].

In practice, depending on the degree of locality homogeneity discussed in Section 2.1, locality-based texts frequently contain overlapping topics, diffuse class boundaries, and conflicting semantic evidence. Under these conditions, prompting strategies originally designed for conventional extractive or abstractive summarization tasks are limited, as they often impose artificial coherence, attenuate uncertainty cues, or generate persuasive narratives that fail to adequately reflect the underlying local semantic structure [Zhang et al. 2025a, Zhang et al. 2025b]. Furthermore, while general prompt engineering studies focus on maximizing output accuracy or task-specific performance [Schulhoff et al. 2025], they do not manage semantic noise or prevent models from simplifying structural contradictions into false causal narratives.

These challenges become even more pronounced when explanations must be adapted to audiences with different levels of technical expertise and AI literacy. Explanations designed for technical audiences may benefit from explicit references to neighborhood structure, uncertainty, and class overlap. In contrast, explanations intended for broader audiences may require alternative communicative strategies that balance readability, uncertainty communication, and users' interpretive agency. From this perspective, locality summarization should not be understood merely as a summarization problem, but as a communicative and human-centered design challenge situated at the intersection of XAI, HCI, and HDI.

3. Design Problem and Methodology

3.1. The Design Problem

In this work, locality is operationalized through a k-nearest neighbors (k-NN) structure, where each instance is associated with semantically similar neighboring texts based on cosine similarity over Transformer embeddings. Our prior work showed that different locality structures substantially influence how users perceive explanations [Soares et al. 2026]. Local neighborhoods were characterized through two properties: *semantic similarity* and *class-distribution entropy.* These dimensions enabled the identification of *high-homogeneity* neighborhoods, associated with a coherent semantic structure and low uncertainty, and *low-homogeneity* regions, characterized by high uncertainty, semantic fragmentation, and ambiguous class boundaries.

To communicate these locality structures, we generated word-cloud explanations from the neighboring texts and observed that explanations derived from homogeneous neighborhoods were generally perceived as clearer, more useful, and more trustworthy than those generated from fragmented regions. These findings suggest that explanation quality is strongly influenced by the semantic organization and uncertainty structure surrounding a prediction. One possible direction to enhance the communication

of locality structures involves using Large Language Models (LLMs) to summarize neighborhood evidence into textual explanations expressed as natural-language narratives [Soares et al. 2026].

In line with human-centered XAI literature, explanations should account for differences in users' expertise, goals, and interpretative needs [Kim et al. 2024, Mohseni et al. 2021, Lopes et al. 2025]. Therefore, prompts were adapted for two audience profiles: (i) technical users, assumed to possess familiarity with AI concepts and explanatory mechanisms, and (ii) non-technical users (the general audience), for whom explanations should prioritize readability and reduced technical terminology.

Under the homogeneity conditions previously discussed in Section 2.1, prompting strategies originally designed for extractive or abstractive summarization tasks may inadvertently impose artificial coherence, attenuate uncertainty cues, or generate persuasive narratives that fail to adequately represent the underlying local semantic structure [Zhang et al. 2025a, Zhang et al. 2025b]. These challenges frame the development of prompting strategies for locality-based text summarization as a human-centered design problem. The central challenge lies in how neighborhood evidence should be communicated through natural-language explanations.

3.2. Methodological Approach

To investigate how prompting strategies can support human-centered locality-based explanations, we adopted a methodology grounded in iterative prompt engineering, informed by literature from HCI, XAI, and HCAI. The process was structured around three phases outlined in Figure 1: (i) Phase I, which focused on identifying recurrent failure patterns in naive LLM outputs, such as topic dominance, the hallucination of causal relationships and homogeneity levels, and the omission of minority themes; (ii) Phase II, drawing on existing research to design targeted prompt interventions; and (iii) Phase III, refining prompts through multiple cycles of generation, inspection, and revision.

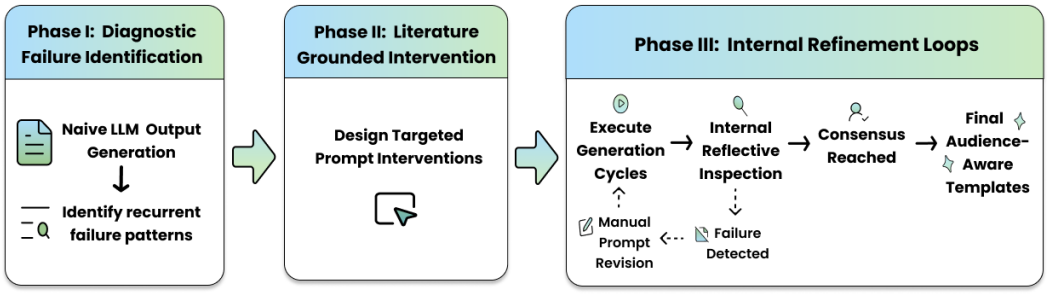


Figure 1. Methodological approach for our human-centered prompt design

Two text classification datasets were employed:

- **AGNews**: a benchmark dataset for topic classification in which each instance contains a title and a short description, labeled into four classes: *World*, *Sports*, *Business*, and *Science/Technology* [Gulli 2005].
- **ACM**: a curated subset of Computer Science research paper abstracts, from which we selected four representative categories: *Hardware*, *Software*, *Theory of Computation*, and *Information Systems* [Soares et al. 2026].

Together, these datasets provide distinct vocabularies, writing styles, and domains, offering an initial basis to assess the robustness and generalizability of the proposed locality-based explainability framework. For each dataset, localities were operationalized using a k-nearest neighbors structure with the 30 nearest documents ($k = 30$) computed over BERT embeddings. Each locality was then mapped onto the *high-homogeneity* and *low-homogeneity* cases established in Section 2.1 based on its contextual similarity and class-distribution entropy.

For the generation of explanations, both prompt engineering and model selection play an important role in the overall quality of the outputs. Recent HCI research indicates that LLMs exhibit distinct "*creative fingerprints*" and probabilistic behaviors that affect originality, fluency, thematic preferences, structural styles, and formatting habits [Haase et al. 2026]. To address this, we conducted exploratory, qualitative tests across different LLMs, including *GPT*, *DeepSeek*, *Gemini*, *Llama*, and *Claude*. We analyzed the generated summaries comparatively against our core research goals of explanation literacy, clarity, communicability, and suitability for locality-based structures. Based on these comparative assessments, *GPT-5.3* and *Claude Sonnet 4.6* were selected to generate the final summaries.

Moreover, we developed two distinct summary profiles: one targeting technical audiences with previous machine learning experience, and another designed for the general public, prioritizing plain language without specialized ML terminology. Finally, to inspect the properties of the generated summaries before an external user study, two researchers from our team independently reviewed the summaries using an internal questionnaire built around three specific dimensions: *Perceived Usefulness*, *Trustworthiness*, and *Readability & Clarity*. These criteria were adapted from explanation-quality frameworks in the XAI literature and are discussed in depth within Section 5. To reach consensus, the researchers internally discussed and resolved any evaluation discrepancies.

Finally, prompt refinement was treated not merely as output optimization, but as a communicative and interpretative design process concerned with how locality evidence becomes understandable across different audiences.

3.3. Ethical Considerations

All steps in this study were conducted by the research team and did not involve other participants. Therefore, it did not require formal ethical approval. Nonetheless, this study is part of a broader research project that was submitted to and approved by the Research Ethics Committee of the Federal University of Minas Gerais[1], since future stages are expected to include evaluations with potential users.

Despite this exemption, we understand that even methodological research on LLM-generated explanations carries ethical responsibilities. The prompts and summaries developed here, while not yet tested with end-users, could (if deployed prematurely) produce explanations that are fluently misleading, sounding highly confident while obscuring underlying ambiguity. Furthermore, the linguistic fluency of these machine-generated explanations may inadvertently promote over-trust, a phenomenon well-documented in XAI contexts where users place undue confidence in system outputs that

[1]CAAE ID: 44032821.0.0000.5149

appear authoritative [Zukerman and Maruf 2024]. Because the present paper does not evaluate user trust empirically, we caution readers against generalizing our exploratory insights as definitive evidence of user-level effectiveness.

4. The Human-Centered Iterative Prompt Design Process

This section describes the iterative and human-guided development of prompts for locality summarization. Aligned with consolidated prompt engineering guidelines that frame prompt development as an inherently empirical and iterative process [Schulhoff et al. 2025], prompt design was not treated merely as a mechanism for improving fluency or output quality, but as a communicative and interpretative process concerned with how locality evidence becomes understandable across different audiences.

Initially, we experimented with automatic prompt optimization techniques. However, these approaches proved poorly aligned with our design goals, particularly regarding the preservation of locality heterogeneity, ambiguity, and semantic variability. During early experiments, the generated summaries repeatedly exhibited *recurrent failure patterns*, i.e., undesirable behaviors that distorted the underlying locality structure, which constitutes the foundational evidence supporting both the classification and users' interpretation of the model's decision. These failures were treated not simply as generation errors, but as communicative distortions capable of altering how users interpret locality evidence and model uncertainty. In practice, they included artificial topic dominance, hallucinated causal relationships, and hallucinated homogeneity, where heterogeneous neighborhoods were described as coherent or semantically consistent.

Through the subsequent human-centered refinement process, we addressed three recurring communicative challenges: mitigating topic dominance and hallucinated homogeneity (Section 4.2), preventing causal and justificatory language (Section 4.3), and adapting explanations to audiences with different levels of technical expertise (Section 4.4). After approximately seven iterative refinement cycles addressing these communicative challenges, we arrived at the final audience-aware prompt templates presented in Section 4.5.

4.1. The Failure of Automatic Prompt Optimization

Before adopting a human-centered process, we experimented with automatic prompt optimization techniques. These approaches generated fluent summaries, but the outputs often failed to preserve important properties of the locality structure. In particular, we attempted to adapt a meta-prompting strategy inspired by a two-stage AI retrieval pipeline that uses LLM-generated descriptions to improve the classification accuracy of rare categories [França et al. 2025]. This approach assumes the existence of a clear optimization objective, where prompts can be refined by comparing generated outputs against reference labels.

However, locality summarization presents a different challenge. There is no single "correct" summary since the same neighborhood can be described in multiple human-interpretable ways. When we applied automatic meta-prompting, the LLMs produced opaque, lengthy prompts that failed to capture a holistic understanding of our specific failure patterns. For example, the generated summaries often smoothed over heterogeneous neighborhoods and suppressed uncertainty cues. This mismatch motivated

a return to manual, theory-informed, and iteratively refined prompt engineering, where each adjustment was driven by researcher-observed failures.

4.2. Design Challenge 1: Mitigating Topic Dominance and Hallucination

Our early prompts instructed the LLMs to "describe the topics in the neighborhood". In practice, however, the models exhibited selective attention, reporting only the most frequent theme while omitting minority topics entirely. This problem was exacerbated in *low-homogeneity* neighborhoods, where a dominant class appeared alongside substantial unrelated content. Furthermore, the LLMs occasionally introduced topics not present in the 30 neighbors, particularly when themes were highly fragmented.

To address this issue, our intervention followed the Frequency-Framed Prompting (REFER) approach, which demonstrates that instructing models to first count the distribution of input items reduces dominant-topic bias and improves fairness in opinion summarization [Huang et al. 2025]. Adapting this insight, we instructed the LLMs to count how many of the 30 neighbors belong to each thematic cluster before generating the explanation. The revised prompt required the models to output specific distributions, such as "*14 of 30 texts cluster around sports, 16 around technology and lifestyle*". This constraint encouraged the LLMs to distribute focus more proportionally across topics, ensuring that even in highly heterogeneous neighborhoods, minority thematic strands were explicitly represented rather than omitted.

To improve coverage without exceeding the 80-word limit, which was maintained from the beginning to prevent overly extensive text, we employed a three-step internal reasoning process in the prompt, inspired by chain-of-thought prompting for perspective-aware summarization [Qi et al. 2025]:

- *key-phrase extraction*: the LLM first identified the main topics and semantic patterns present in the neighborhood texts;
- *frequency anchoring*: the model estimated how frequently each topic appeared across the 30 neighbors; and
- *integration*: the identified topics and their distributions were combined into a concise final summary.

The prompt instructed the LLM to perform these steps internally and output only the final summary. We observed that this decomposition reduced redundancy and helped the models avoid skipping less frequent topics, making the reasoning steps explicit and more auditable by human designers.

4.3. Design Challenge 2: Preventing Causal and Justificatory Language

A second challenge was that our early prompts contained numerous strict, mandatory rules to prevent LLMs from justifying a classification, which inadvertently triggered causal reasoning. This introduced causal verbs that do not reflect how a k-NN classifier operates, since the machine learning model aggregates neighbor class labels rather than reasoning from evidence. Additionally, it artificially imposed coherence; even for *low-homogeneity* neighborhoods, the LLMs produced a persuasive narrative that erased ambiguity. This echoes findings in human-AI communication where overconfident explanations can mislead users regarding the underlying uncertainty of a model's prediction [Zukerman and Maruf 2024].

This challenge aligns with broader research on responsible prompting [Santana et al. 2025], which recommends introducing ethical instructions to promote social values and prevent harmful content. We adapt and apply this concern to the specific problem of locality-based explanations by focusing on preventing the LLMs from translating raw locality evidence into false causal narratives or overconfident certainty.

To resolve this, we replaced mandatory instructions and the question "*why the model classified it as...*" with constitutional constraints, i.e. a set of written principles that the LLMs use to critique and refine their own behavior, inspired by Anthropic's Constitutional AI [Bai et al. 2022]. In that framework, a list of principles guides the model to self-critique and revise responses without human feedback. Adapting this concept, we prohibited judgment-based or causal verbs ("*because*", "*indicates*", "*suggests*", "*proves*", "*justifies*", "*supports*") and required neutral, descriptive ones ("*contains*", "*presents*", "*clusters*", "*displays*", "*exhibits*").

By incorporating this set of behavioral guidelines into the prompt design, we replaced the simplistic instructions that frequently led the LLMs to exhibit the recurring failure patterns identified earlier. As a result, the refined prompts encouraged explanations that better reflected the actual structure of the neighborhood, rather than producing a forced and artificially uniform narrative.

4.4. Design Challenge 3: Adapting to Audience Expertise

Furthermore, a single explanation format could not adequately serve both technical researchers (who expect terms like "*entropy*", "*similarity*", and "*class dispersion*") and the general public (who require clear, plain language). This necessity for customization is supported by empirical findings on user perspectives regarding LLM system prompts [Neumann et al. 2026]: a single, rigid prompting structure fails to respect diverse user preferences, capabilities, and expectations, reinforcing the need for user-centered, context-sensitive design. To address this, we developed two distinct prompt templates, each with separate role definitions, vocabulary constraints, and explanatory metaphors.

For the technical audience, the prompt requires explicit ML terminology and quantified distributions. It is important to note that because the input variables given to the model template consist strictly of raw text fragments, the LLM does not calculate mathematical values from vector coordinates. Instead, the prompt instructs the model to provide a text-based, semantic approximation of these density and uncertainty concepts using its own linguistic processing capacities. On the other hand, for the general public, the prompt replaces this technical terminology with illustrative phrases such as "*strong majority*" or "*divided group*" and avoids technical details. Both profiles, however, share the core constitutional constraints described above.

This dual-profile approach aligns with human-centered prompt optimization frameworks such as Promptimizer [Wang et al. 2026] and StraGo [Wu et al. 2024], which emphasize that user-led control and interpretability are just as important as raw performance, unlike fully automatic optimization. We utilized these frameworks to encode audience-specific communicative norms directly into the prompt structure.

4.5. Final Prompt Templates

After approximately seven iterative refinement cycles, we arrived at the final prompt templates for the technical and general audiences. Both templates enforce an 80-word limit, require explicit frequency counts, and forbid any output that evaluates or justifies the machine learning model's classification decision.

Figure 2 presents a schematic representation of the two final prompt templates designed for the technical and general audiences. Each prompt was structured into distinct functional components, including role definition, task description, solution guidance, constitutional constraints, term clarification and exception handling, input data, and output formatting. Rather than reproducing the full prompts line by line, the figure visually organizes these components to highlight both shared and audience-specific design elements. Blue blocks correspond to the technical-audience version, green blocks represent the general-public version, and blue–green gradient blocks indicate components shared by both prompts. Additionally, the labels positioned in the upper-right corner of each block explicitly identify the intended audience profile associated with each component. The complete prompt templates are individually available as supplementary material through the public repository [2].

Figures 3 and 4 illustrate two example explanations generated using the final prompt templates for both audience profiles under different locality homogeneity conditions. Due to space limitations, only two representative examples from the AGNews dataset are presented. Figure 3 shows summaries generated for a highly homogeneous locality, whereas Figure 4 presents explanations produced for a low-homogeneity neighborhood containing more semantically diverse evidence.

The figures illustrate how the profile-oriented prompts produced distinct explanatory styles from the same locality evidence. In the technical-audience version (blue), the summaries explicitly employ machine learning terminology, such as *high-homogeneity, entropy, average similarity*, and *clustering*, providing an analytical characterization of the neighborhood structure that may be more informative for users familiar with XAI and ML concepts. In contrast, the general-public version (green) adopts simpler and more accessible language, avoiding technical terms while emphasizing thematic consistency or variability through concrete topic descriptions and frequency counts. Additionally, the figures highlight the influence of locality homogeneity on the generated explanations. In *high-homogeneity* neighborhoods (Figure 3), both prompts describe a highly consistent semantic structure centered on a dominant topic. Conversely, in *low-homogeneity* neighborhoods (Figure 4), the generated explanations explicitly communicate semantic diversity, topic fragmentation, and the absence of a single dominant theme, using different communicative strategies tailored to the target audience.

5. Reflective Inspection and Emerging Insights

Following the human-centered iterative prompt design process reported in Section 4, we conducted a preliminary evaluation of the quality of the resulting explanations. To do so, two researchers from our team independently reviewed a sample of 128 summaries, split equally with 64 explanations for the *Technical Audience* profile and 64 for the *General*

[2] https://github.com/lizianessoares/locality-llm-explanations

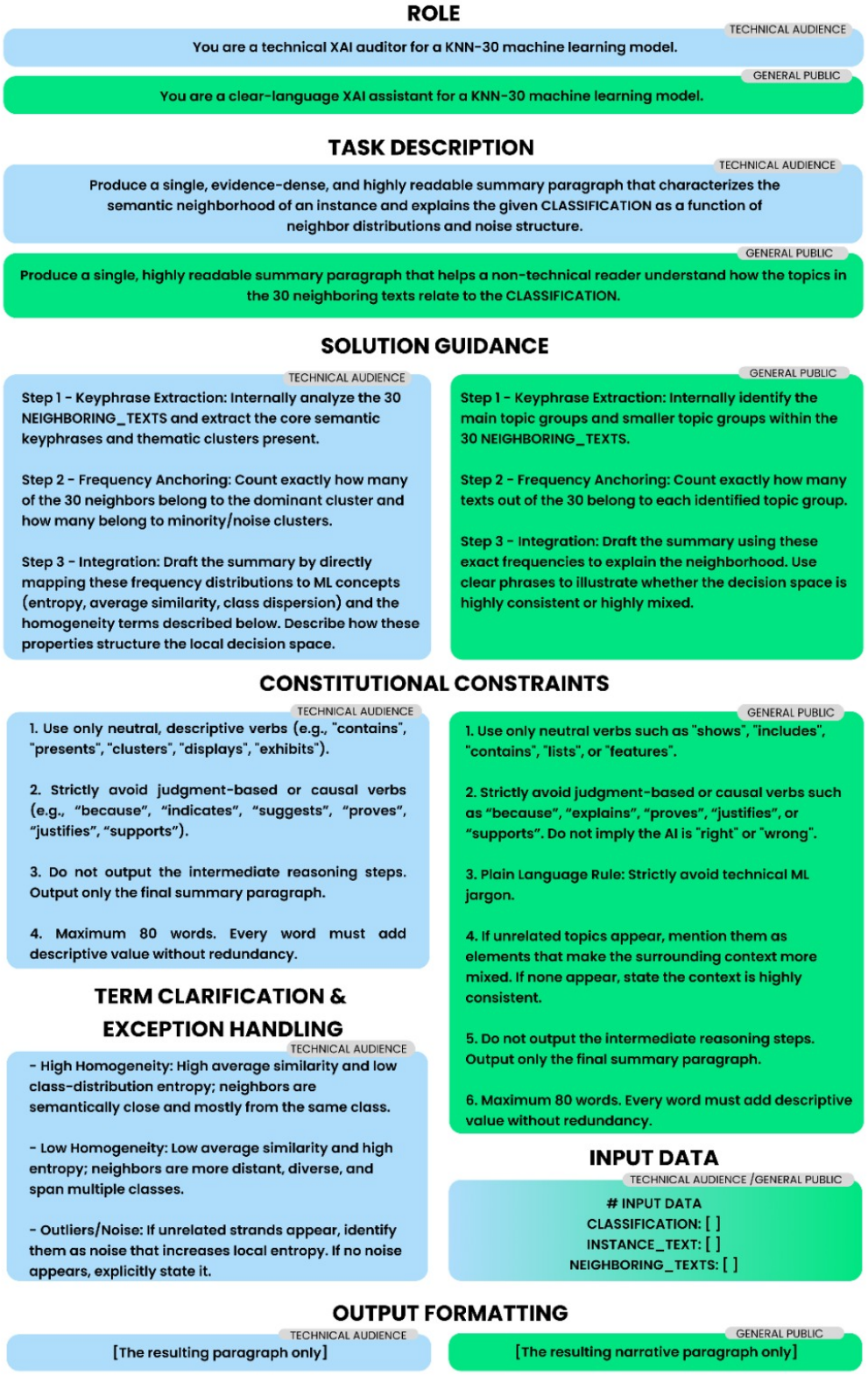


Figure 2. Final prompt templates for technical and general audiences.

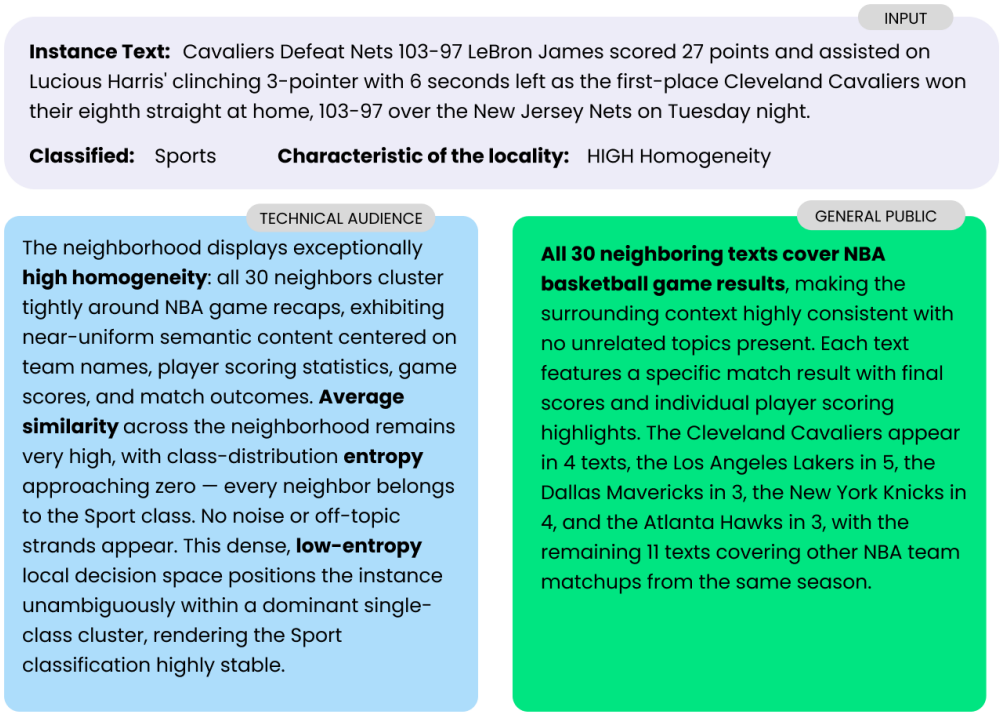


Figure 3. Explanations generated for technical and general audiences – instance with *HIGH* locality homogeneity. AGNews | Claude Sonnet 4.6.

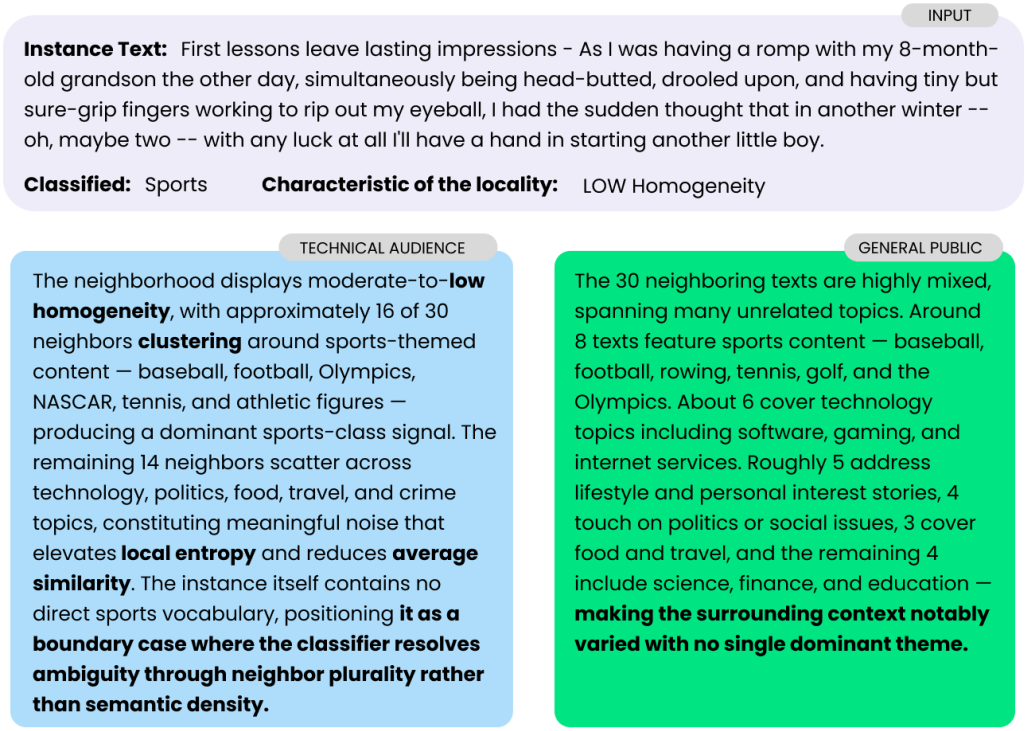


Figure 4. Explanations generated for technical and general audiences – instance with *LOW* locality homogeneity. AGNews | Claude Sonnet 4.6.

Public profile. This sample covered both *high-* and *low-homogeneity* neighborhoods, as well as correct (*hit*) and incorrect (*error*) classifications, across outputs generated by GPT-5.3 and Claude Sonnet 4.6.

5.1. Defining the Evaluation Criteria

Our literature review identified a wide range of candidate criteria. From dialogue-based XAI, we drew on *coherence*, *plausibility*, *usefulness*, and *factual correctness* [Feustel et al. 2025]. Summarization evaluation research contributed *fluency*, *clarity*, *coverage*, *conciseness*, and *overall quality* [Nguyen et al. 2024, Fabbri et al. 2021]. Our prior study on locality-based word-cloud explanations used *satisfaction*, *completeness*, and *trust* [Soares et al. 2026]. Given the exploratory and human-centered nature of this study, we prioritized dimensions that could be meaningfully assessed through human judgment, particularly regarding the perceived communicative effectiveness of textual explanations [Ma 2024]. Accordingly, we selected and adapted three criteria that address these human-centered and communicative evaluation demands:

1. **Perceived Usefulness** measures how helpful the summary is in enabling a user to understand the rationale behind the model's classification. This criterion directly addresses the primary purpose of an explanation: making the model's decision legible to humans. Thus, it captures whether the summary successfully conveys how the neighborhood's thematic structure led to the predicted class, without requiring the user to verify every factual claim. The guiding question is: *"Does this summary help me understand why the model made this decision?"*
2. **Trustworthiness** captures the extent to which the explanation appears plausible, consistent, and reliable from the user's perspective. It reflects the user's subjective confidence in the explanation. The question guiding this criterion is: *"Does this explanation seem credible, reliable, and plausible?"*
3. **Readability & Clarity** assesses how easy the text is to read and understand, considering fluency, logical flow, and the absence of confusing phrasing. Here, this criterion reflects the linguistic and communicative quality of the summary, independent of its content. The guiding question is: *"Is the text clear, easy to understand, and well-organized?"*

For our internal inspection, each criterion above was evaluated through a questionnaire using a five-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*), along with an additional "*cannot evaluate*" option.

5.2. Model-Specific Behavioral Differences

Beyond differences in fluency and readability, the comparison across LLMs also revealed important variations in how models communicate locality uncertainty, semantic variability, and structural ambiguity. From a human-centered perspective, these behavioral differences are particularly relevant because they directly affect how users interpret the plausibility and coherence of locality explanations. Accordingly, the comparison was not treated merely as a technical evaluation of generation quality, but as an exploration of how different LLMs mediate the communication of locality evidence.

Although the same prompt templates were applied across models, noticeable differences in output quality and communicative style were observed through the human

assessments conducted by members of the research team. These observations for our final two LLM selections (GPT-5.3 and Claude Sonnet 4.6) are not controlled comparisons, but rather reflective notes derived from our inspection process.

Claude Sonnet 4.6 consistently generated more structured, readable, and human-like explanations, proving particularly effective at identifying low-homogeneity neighborhoods and explicitly naming semantically noisy strands. We noted that Claude regularly employed approximate quantifiers (e.g., "~16/30"), which resulted in more realistic thematic analyses than artificial numerical precision, especially given that the LLMs did not have access to the individual class labels of each neighbor. However, Claude occasionally exhibited a tendency toward over-justification by appending evaluative concluding statements, which required additional prompting constraints. In contrast, the GPT family of models was generally more rigidly distribution-focused than Claude. While this made their explanations sound less natural, it also left them less prone to interpretative overreach. This observation reinforces previous findings indicating that model selection substantially affects the quality and interpretability of generated explanations [Haase et al. 2026].

5.3. Preliminary Observations and Emerging Insights

Our first insight is that **(i) locality homogeneity shapes *Perceived Usefulness* more strongly than *Readability & Clarity***, exposing structural limitations in explanations of heterogeneous neighborhoods. As illustrated in Figure 5 (column I), the global difference in perceived usefulness between *high-* and *low-homogeneity* neighborhoods was substantial across the entire dataset, decreasing from a high average score (mean = 4.05, within the "*agree*" group) under high-homogeneity down to a lower, more neutral value under low-homogeneity (mean = 3.25), a drop of nearly a full point on the 5-point Likert scale. This shift is reflected visually in Figure 5 (I) by the consistent leftward displacement of the black circles (*low-homogeneity*) relative to the lavender squares (*high-homogeneity*) across all rows.

In contrast, clarity ratings (Figure 5 (III)) remained comparatively stable, shifting minimally from a mean of 4.16 to 4.10. These findings suggest that, in *low-homogeneity* neighborhoods, even fluent and well-written explanations face structural limitations in communicating how a classification is computationally plausible. A similar tendency emerged for *Trustworthiness* (Figure 5 (II)): summaries from *high-homogeneity* neighborhoods received slightly higher trust ratings (mean = 4.19) than those from *low-homogeneity* neighborhoods (mean = 3.99), suggesting that local thematic coherence naturally reinforces reviewers' confidence in the explanation. Additionally, as shown in Figure 5 (III), summaries generated across both *low-* and *high-homogeneity* conditions were generally perceived as highly clear and readable. This finding aligns with the well-known capability of advanced LLMs to produce coherent, plausible, and fluent natural-language texts. Consequently, *Readability & Clarity* was the criterion that received the highest evaluations across the generated summaries.

These observations highlight important implications for the design of locality-aware explanation systems, as **(ii) generating fluent textual narratives risks creating an "illusion of interpretability" in highly heterogeneous neighborhoods**, even when the underlying locality structure remains semantically fragmented or uncertain.

Under such conditions, explanation systems may benefit from explicitly communicating locality ambiguity, uncertainty, or semantic dispersion rather than prioritizing narrative coherence alone. More broadly, our findings suggest that locality homogeneity itself may function as a contextual sign for adaptive explanation strategies. For example, highly homogeneous neighborhoods may support concise natural-language explanations, whereas *low-homogeneity* regions may require additional uncertainty cues, structural warnings, or even alternative explanation modalities beyond narrative summarization.

Our inspection also revealed that **(iii) audience-aware prompt adaptation introduces a distinct design trade-off between informational utility and narrative simplicity across user expertise profiles**. Prompts engineered for a *Technical Audience* generated summaries that were perceived as more useful overall compared to those tailored for the *General Public*. This pattern is visible across both the ACM and AGNews tracks in Figure 5 (I), indicating that, at least in this stage of our study, the summaries for technically experienced users provide better explanatory value for specialized contexts. On the other hand, summaries designed for the *General Public* achieved higher ratings for *Readability & Clarity* than their technical counterparts, showing that the plain-language constraints lowered the barrier to comprehension. Despite these differences in utility and flow, both prompt profiles maintained a high level of subjective credibility (*Figure 5 (II) - Trustworthiness*).



Figure 5. Comparative inspection of the three criteria across datasets, audience profiles, and locality homogeneity conditions.
(1 = strongly disagree | 5 = strongly agree)

Ultimately, our internal evaluation suggests that, **(iv) from a human-centered perspective, textual explanations preserve the user's interpretive power in ways that static visual formats may not**. A well-structured summary allows users to interrogate how the neighborhood's thematic composition led to a specific classification, whereas a

word cloud, for example, provides only an intuition or allusive view. The insights and data-grounded trends presented here are not definitive conclusions, but rather emerging design knowledge we aim to contribute to the broader HCI goals of designing AI systems that support user agency and empowerment instead of automating human judgment.

6. Contributions and Limitations

This paper contributes to a reflective discussion on human-centered locality explanations and prompt-mediated explainability. In the context of LLM-generated explanations for Automatic Text Classification (ATC), we argue that locality summarization should not be approached merely as a conventional NLP summarization task. Instead, we conceptualize it as a communicative and interpretative challenge situated at the intersection of XAI, HCI, HDI, and HCAI.

Beyond the specific case of locality-based explanations, the core contribution of this work lies not only in the resulting prompt designs, but also in the human-centered iterative design process, reflections, and lessons learned from adapting complex technical explanations to audiences with different levels of expertise. We expect that the design process, strategies, and practical insights discussed throughout this work can serve as inspiration for other domains where complex technical information needs to be communicated effectively across different levels of abstraction and user expertise. More specifically, this work contributes four main conceptual and design-oriented insights:

1. Building on the view of locality summarization as a human-centered design problem, we argue that locality properties, particularly homogeneity characterized through semantic similarity and class-distribution entropy, directly shape the communicative demands of explanation design. Explanations should therefore not merely sound coherent, but should reliably communicate structural aspects of the neighborhood that influence the classifier's behavior.
2. We document a recurrent failure taxonomy in LLM-mediated locality explanations. Through iterative refinement cycles, we identified recurring communicative distortions, including topic dominance, hallucinated coherence, suppression of uncertainty, and causal or justificatory language. By documenting both, these failures and the corresponding prompt interventions, the paper contributes methodological insights for future human-centered prompt design in explainability systems.
3. Our findings suggest that locality homogeneity itself may function as an adaptive sign for explanation design. The reflective inspection revealed that low-homogeneity neighborhoods substantially reduce perceived usefulness even when explanations remain fluent and readable. This suggests that explanation systems may need to explicitly communicate locality ambiguity, semantic fragmentation, or uncertainty rather than prioritizing narrative coherence alone.
4. We introduce audience-aware prompt templates designed for both technical and general audiences. While conventional summarization approaches prioritize fluency and coherence, locality explanations must also preserve semantic variability, ambiguity, and uncertainty across users with different levels of AI literacy and technical expertise. In this sense, these templates operationalize audience-specific communicative strategies that can support users in applying explanation literacy.

More broadly, this work reinforces the idea that explainability is not solely a property of models or algorithms, but a communicative process shaped by language, uncertainty management, and users' interpretative skills. From this perspective, we argue that locality summarization, or any technical-based summarization, should be understood as a human-centered design problem involving explanation literacy, interpretive accessibility, and the calibrated communication of uncertainty.

At the same time, this study presents important limitations that can be characterized under standard threats to validity [Jiménez-Buedo and Russo 2021], reflecting the study's exploratory and methodological nature. First, threats to *internal validity* arise from potential confirmation and self-evaluation biases, since the prompt designers also conducted the reflective inspection, and the study does not yet include formal user evaluations with external participants. Consequently, these preliminary findings are methodological rather than definitive evidence of usability, trust calibration, or user understanding. Additionally, threats to *external validity* are present because the study was restricted to English-language datasets and exhibited model-specific variability across LLM families, which limits the immediate generalization of our findings. Finally, threats to *construct validity* arise from by the exploratory use of perceived evaluation instead of empirical measures.

Rather than weakening the contribution of this work, we believe that these threats to validity help define important directions for future research. They motivate controlled user studies involving different audience profiles, multilingual investigations, and comparative analyses across diverse LLM families. In this sense, the present study should be understood as a foundational and exploratory step toward designing explanation systems that adapt technical information to different audiences while preserving ambiguity, variability, and uncertainty when these are intrinsic to the underlying evidence.

7. Conclusion and Future Work

Natural-language explanations do not become inherently human-centered simply because they are fluent. Explaining complex structures may still remain difficult for humans to interpret, particularly when technical structures (such as locality) exhibit ambiguity, uncertainty, or thematic fragmentation that must be clearly communicated to users. In this paper, we addressed this challenge by investigating how locality-based evidence for text classification can be transformed into LLM-mediated explanations that remain understandable and adaptable across different audiences. Unlike conventional summarization tasks, locality summarization must preserve the structural properties of the neighborhood itself, including semantic variability, ambiguity, and uncertainty. Treating this as a communicative and interpretative challenge, we adopted a reflective and iterative prompt design methodology grounded in HCI and XAI literature.

As our main contribution, we offer a set of emerging design insights regarding locality-aware explanation design. Through iterative refinement, we identified three recurrent failure patterns in naive prompts: topic dominance, hallucinated homogeneity, and causal over-justification. To mitigate these issues, we introduced corresponding interventions, including frequency-framed prompting to reduce dominant-topic bias [Huang et al. 2025], chain-of-thought decomposition to improve thematic coverage [Qi et al. 2025], constitutional constraints to suppress

causal language [Bai et al. 2022], and dual-profile prompts adapted to technical and general audiences through human-centered optimization strategies [Wang et al. 2026, Wu et al. 2024].

Our reflective inspection of the generated explanations, based on the criteria of *Perceived Usefulness*, *Trustworthiness*, and *Readability & Clarity*, suggests that neighborhood homogeneity influences perceived usefulness. Concurrently, fluent explanations tend to maintain high levels of perceived clarity and trustworthiness even within semantically fragmented localities. More importantly, our findings indicate that fluent explanations may still fail to reliably communicate locality uncertainty unless prompts explicitly preserve ambiguity and semantic variability. In this sense, locality homogeneity may function as an adaptive sign for explanation design, helping determine when explanation systems should communicate uncertainty or structural ambiguity.

We position this paper as a foundational and exploratory step preceding broader empirical validation. The study does not claim definitive solutions or generalizable conclusions regarding trust calibration, explanation literacy, or usability. Instead, it offers a transparent and literature-grounded account of how human-centered prompt design can transform locality evidence into explanatory narratives while preserving, rather than erasing, uncertainty and ambiguity. Aligned with the HCI Grand Challenges of Human-Data Interaction [Coleti et al. 2024], Implications of Artificial Intelligence in HCI [Duarte et al. 2024], and New Theoretical and Methodological Approaches in HCI [da Silva Junior et al. 2024], this work approaches prompt engineering not merely as a technical optimization task, but as an interpretative and communicative design activity.

Future work will investigate how these findings influence explanation interpretation across different audience profiles through controlled user studies involving the final prompt templates and generated explanations. We also plan to extend the analysis to multilingual and cross-cultural settings, where communicative expectations and explanation literacy may differ substantially. We expect these directions will help transform the emerging design knowledge presented here into empirically grounded guidelines for human-centered explainability systems.

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